Six specialised governance systems. Not one general prompt.

Brand governance, built the way brand leads build it.

LoomShield reviews content before publication across tone, terminology, messaging, visual & lockup consistency, accessibility, and promotional claims — explaining every issue and suggesting surgical fixes that preserve the creator's original purpose.

Construction	**Two local engines**
Verification	**Rules, facts & lockups**
API keys	**None — 100% on-device**

PROJECT NAME	TEAM NAME	PARTICIPANT
LoomShield	**isaac.incognito007**	**Isaac Shiloah Herbert V**

1. My Understanding of the Given Problem Statement

01 / PROBLEM

Growing organisations publish content simultaneously through internal employees, regional teams, external agencies, and AI drafting tools. Just as a general image model produces something that merely looks logo-shaped without respecting typographic weight, clear-space ratios, or lockup geometry, unguided content creation produces material that looks finished but quietly breaks brand identity and factual accuracy across six areas:

1. Unsupported Claims

Unverified superlatives (*"#1 fastest"*, *"100% guaranteed ROI"*) and outdated product numbers get published without factual backing, creating legal and reputational risk.

2. Tone & Voice Drift

External agencies and AI drafting tools often adopt alarmist, overly casual, or off-brand tone that conflicts with the organisation's character.

3. Terminology Errors

Deprecated product names, prohibited buzzwords, and inconsistent capitalization weaken brand consistency across channels.

4. Off-Brand Visuals & Lockups

Visuals use unauthorized hex colors (high CIE ΔE deviation), broken typography scales, or violated logo clear-space and lockup rules.

5. Accessibility Gaps

Low text contrast (<4.5:1), missing image alt-text, overly complex prose, and skipped heading levels violate WCAG 2.1 AA standards.

6. Unclear Guidelines

Existing checkers either reject drafts without citing the rule, overwrite the writer's voice, or fail to show leadership which brand guidelines are unclear.

Which Specific Part of the Problem I Intend to Solve

Static PDF style guides are ignored under tight deadlines, while generic cloud AI wrappers incur ongoing API costs, risk leaking confidential pre-launch materials to third-party servers, and rewrite copy without citing rules. I am designing **LoomShield** to solve pre-publication brand governance as a constraint-driven system: checking multi-format content locally against approved guidelines, facts, and visual specs, explaining each issue with an exact source citation, suggesting surgical fixes that preserve the creator's purpose, supporting role-based approval workflows, and highlighting where the brand guidelines themselves are ambiguous.

2. The Users Affected by the Problem

02 / USERS

Organisational Setting: Growing B2B/B2C companies, design-led brands, regulated industries (fintech, healthcare, SaaS), and multi-agency marketing teams where drafts move from creators to reviewers before publication.

User Group	Role in Content Workflow	Current Challenges	How LoomShield Helps Them
Content Creators & Designers *(Marketers, Copywriters, Agencies, AI-Assisted Creators)*	Draft blog posts, launch emails, landing pages (`.html/.docx`), and visual lockups/banners (`.png/.svg`).	Cannot memorize every approved metric, hex code, or clear-space rule; frustrated when reviewers reject drafts late or when AI tools overwrite their voice.	Clicking **Shield Your Loom → LoomShield** runs an instant check with exact rule citations and 1-click surgical fixes that keep their purpose intact.
Reviewers & Managers *(Brand Managers, Creative Directors & Compliance Leads)*	Inspect drafts, verify promotional claims and visual lockups, leave comments, and approve or request changes.	Spend hours manually cross-checking statistics against fact sheets and checking color/clear-space rules across scattered threads.	Automates 6-pillar verification, tracks version history with side-by-side diffs, and logs human override decisions.
Brand Governance Leads *(CMO, Head of Brand, Design Systems Owners)*	Define and maintain brand guidelines, approved facts, terminology rules, typography scales, and symbol/asset libraries.	Lack visibility into repeated mistakes or whether high violation rates happen because a guideline is vaguely written.	Uses a **Guideline Ambiguity Radar** to flag rules with high violation and override rates so unclear guidelines can be fixed.

3. My Proposed Solution: LoomShield

03 / SOLUTION

My proposed solution is **LoomShield**—an intelligent, local-first brand-governance platform launched via the **Shield Your Loom → LoomShield** action menu. Rather than asking one general cloud prompt to critique a draft, LoomShield evaluates text, documents, and visual assets against concrete constraints stored in the organisation's **Brand Vault** (tone rules, terminology lexicon, verified fact sheet, typography/symbol clear-space rules, and approved hex palette). For every issue found, it explains why it was flagged, cites the exact rule or fact source, proposes a surgical correction, and routes the draft through a role-based review workflow.

Local-First, Cloud-Optional Architecture (100% Zero-Key Dual On-Device Engine)

LoomShield does not require users to pay for external AI API services, reducing operational costs to $0 while giving organisations full privacy and control over confidential materials. My design follows four strict rules:

1. **Process ordinary requests with deterministic rules and embedded NLP:** All everyday inspections across tone, terminology, messaging, promotional claims, visual/lockup consistency, and accessibility run locally on-device using deterministic rules paired with embedded machine-learning NLP (TF-IDF vectorization, cosine similarity fact retrieval, tone polarity scoring, Flesch-Kincaid readability, CIE ΔE color distance, clear-space boundary checks, and WCAG 2.1 contrast math).
2. **Use a cloud AI API only for requests that genuinely need advanced language reasoning:** An optional cloud escalation gate is used only when a user explicitly requests complex multi-paragraph rhetorical restructuring and confidential mode is disabled.
3. **If no API key is configured, keep all local features working and explain which advanced features are unavailable:** With zero API keys configured, 100% of the core 6-pillar inspections, surgical suggestions, visual heatmaps, and ambiguity insights work locally, while the UI clearly states which optional cloud feature is inactive.
4. **Keep any cloud API key on a secure backend, not in public frontend code:** Any optional API key is stored strictly in backend environment variables (`.env`) and never exposed in client-side code.

Engine Configuration Matrix

Your Requirement	API Key?	NLP?	Role in My Architecture
Local deterministic engine only	**No**	Basic, rule-based NLP	**Tier 1 (On-Device):** Lexicon & capitalization rules, regex claim patterns, WCAG 2.1 contrast math, and CIE ΔE color/clear-space checks.
Local engine + embedded NLP model	**No**	**Yes** including machine-learning NLP	**Tier 2 (On-Device):** Scikit-Learn TF-IDF / Cosine RAG Fact Grounder, Tone Polarity Classifier, and Surgical Diff Synthesizer.
Local engine + cloud AI fallback	Usually yes	**Yes** locally and potentially in the cloud	**Tier 3 (Optional Gate):** Backend-only cloud AI fallback for heavy cross-document restructuring when permitted.
Two local engines, both running on-device	**No**	**Yes** **if the engines include NLP**	**LoomShield Default Mode: Runs Tier 1 (Deterministic) + Tier 2 (Embedded ML/NLP) together on-device with zero API cost.**

Key Benefits of My Zero-Key On-Device Approach

- **Zero External API Cost:** Teams can run unlimited pre-publication reviews without per-token API bills or paid subscriptions.
- **Confidentiality & Data Control:** Embargoed press releases, unannounced brand marks, and internal assets stay 100% on the local machine.
- **Constructed Precision Over Sampled Guesswork:** Every flag is reasoned against explicit Brand Vault constraints in under 100ms with zero hallucinations.

4. The Complete Product Workflow

04 / WORKFLOW

Five stages, each handled by a dedicated module built for that job:

STAGE 1 • VAULT
Define Constraints
Brand Admins import tone guidelines, terminology rules, a verified fact sheet with citations, approved hex colors, and lockup clear-space rules.

STAGE 2 • INGEST
Shield Your Loom
Creators draft copy or upload documents (`.html` , `.docx` , `.md`) and visuals (`.png` , `.svg`) and click **Shield Your Loom → LoomShield**.

STAGE 3 • AUDIT
Dual Local Check
Tier 1 (Deterministic) and Tier 2 (Embedded NLP) inspect all 6 pillars simultaneously against the Brand Vault constraints.

STAGE 4 • REFINE
Surgical Corrections
Each flag cites the exact rule or fact and suggests an inline diff. Creators can **Accept**, **Modify**, or **Dismiss with Reason**.

STAGE 5 • GOVERN
Approve & Learn
Reviewers comment, inspect version diffs, and approve drafts. Override patterns feed the **Guideline Ambiguity Radar**.

5. The Main Features I Intend to Build

05 / FEATURES

Feature Module	Planned Capabilities
1. Structured Brand Vault & Asset Constraints	Central store for tone rules, terminology lexicon (preferred vs. prohibited terms), verified fact sheet (approved statistics, pricing, SLAs with source citations), typography/lockup rules, and approved brand hex colors.
2. "Shield Your Loom → LoomShield" Multi-Format Reviewer	One-click governance launcher that accepts marketing copy, uploaded documents (`.html` , `.docx` , `.md` , `.txt`), and campaign visuals/logos (`.png` , `.jpg` , `.svg`) along with contrast and alt-text metadata.
3. 6-Pillar Dual On-Device Hybrid Engine	Detects issues across: **(a) Tone** (polarity and alarmist phrasing), **(b) Terminology** (banned words and capitalization), **(c) Messaging** (alignment with brand pillars), **(d) Promotional Claims** (superlative regex + TF-IDF cosine verification against verified facts), **(e) Visual & Lockup Consistency** (CIE ΔE color distance and clear-space margins), and **(f) Accessibility** (WCAG 2.1 AA contrast, optical sizing/readability, alt-text, heading hierarchy).
4. Explainable Citation Cards & Surgical Rewriter	Displays the exact violated `Rule ID` or `Fact ID` , quotes the official source, explains why it was flagged, and proposes a surgical replacement that preserves the creator's surrounding sentence.
5. Role-Based Approval Workflow & Revision History	Supports `Creator` , `Reviewer` , and `Brand Admin` roles with status transitions (`Draft → In Review → Changes Requested → Approved`), inline comments, and version history diffs.
6. Confidentiality Gate & Engine Transparency	Guarantees 100% on-device processing for confidential drafts and labels every finding with its engine source (`Tier-1 Deterministic` vs. `Tier-2 Embedded NLP`).
7. Guideline Ambiguity Radar & Clarity Rewriter	Correlates violation counts with human override reasons to detect unclear brand guidelines, and uses the Two-Tier Hybrid Engine to draft clearer rule definitions with Do/Don't examples.

6. The Technology Stack I Plan to Use

06 / STACK & ARCHITECTURE

Layer	Planned Technologies	Why I Selected This Stack
Frontend UI	React / TypeScript, HTML5, CSS3	Enables a responsive multi-pane review studio with inline span highlights, side-by-side surgical diffs, color/lockup inspectors, and role switching.
Backend API	Python 3.11, FastAPI, Pydantic v2	Fast asynchronous API with strict schema validation ensuring every finding includes a verified citation, character offsets, and confidence score.
Tier 1: Local Deterministic Engine	Regex/AST Scanner, WCAG 2.1 Luminance Math, Pillow + NumPy CIE ΔE Analyzer	Runs exact terminology checks, statistical claim extraction, Flesch-Kincaid readability, WCAG contrast ratios, and dominant image color/clear-space checks in under 30ms.
Tier 2: Embedded On-Device NLP Engine	Scikit-Learn TF-IDF Vectorizer, Cosine Similarity RAG, Polarity Classifier	Performs on-device semantic retrieval against the Verified Fact Sheet to catch exaggerated claims and generates surgical rewrites without needing an API key.
Database & Storage	SQLite 3	Zero-configuration relational storage for brand guidelines, verified facts, multi-version submissions, review comments, and override telemetry.

SYSTEM ARCHITECTURE: LOCAL-FIRST DUAL ON-DEVICE HYBRID ENGINE (CLOUD-OPTIONAL) SHIELD YOUR LOOM → LOOMSHIELD

1. Multi-Format Input

- Copy & Markdown
- Docs (`.html` / `.docx`)
- Visuals (`.png` / `.svg`)

Confidentiality Gate: On-Device

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2. Dual On-Device Hybrid Engine (Zero API Key)

Tier 1: Deterministic

- Lexicon & Case Rules
- Claim & Stat Regex
- WCAG 2.1 Contrast Math
- Palette ΔE & Clear Space

Tier 2: Embedded NLP

- TF-IDF Cosine Fact RAG
- Tone Polarity Vectors
- Surgical Diff Rewriter
- Rule Ambiguity Clarifier

Optional Tier 3: Backend Cloud AI Fallback (Inactive When Zero-Key)

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3. SQLite Brand Vault

- Tone & Style Rules
- Verified Fact Sheet
- Terminology Lexicon
- Hex & Lockup Specs

Citations: RULE & FACT ID

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4. Review & Radar

- Inline Surgical Diffs
- Accept / Override Control
- Role Approval Workflow
- Ambiguity Radar Insights

SHA-256 Audit Sign-Off

Most Technically Challenging Part & How My Planned Architecture Solves It

Technical Challenge: Detecting unsupported promotional claims (for example, *"cuts settlement time by 95%"*) and suggesting corrections that preserve the writer's voice without hallucinating facts or relying on paid external LLM APIs.

My Planned Solution: I combine **Tier 1 (Deterministic Regex Span Extraction)** with **Tier 2 (Embedded TF-IDF Cosine Similarity RAG)**. Tier 1 locates statistical claims and superlatives with exact character offsets `[start, end]`. Tier 2 vectorizes the claim context and measures cosine similarity against the **Verified Fact Sheet**. When a draft exaggerates a number (such as claiming *"95% faster"* when `FACT-101` verifies *"up to 64% faster"*), LoomShield cites `FACT-101` and swaps only the inaccurate clause while keeping the rest of the creator's sentence untouched.

7. Responsibilities of Each Team Member

07 / TEAM ROLES

I am competing as a solo participant (Team `isaac.incognito007`) and I am responsible for all four engineering areas of the project:

Team Member	Area of Responsibility	Specific Tasks I Will Execute
Isaac Shiloah Herbert V *Solo Team Leader & Developer* `isaac.incognito007@gmail.com`	**1. System Architecture & Data Layer**	Design the SQLite Brand Vault schema (guidelines, verified facts, terminology lexicon, visual/lockup specs) and local-first privacy controls.
	2. Dual On-Device Hybrid NLP & Visual Engine	Build Tier 1 (lexicon, regex claim extractor, WCAG contrast math, Pillow/NumPy CIE ΔE & clear-space analyzer) and Tier 2 (TF-IDF cosine fact grounder, tone classifier, surgical diff synthesizer).
	3. Full-Stack UI & Review Workflow	Build the interactive review studio, the **Shield Your Loom → LoomShield** launcher, span highlighter, surgical diff acceptance/override controls, role-based approval state machine, and Ambiguity Radar view.
	4. Testing, QA & Documentation	Create realistic multi-brand test scenarios, run automated API and accessibility checks, and prepare demonstrations and documentation.

8. My Development Plan Across All Three Rounds

08 / ROADMAP

Round	Planned Development Milestones	Key Outcomes
Round 1: The Plan	· Define problem scope, user personas, and the 6 detection pillars. · Specify the Local-First Dual On-Device Hybrid Engine architecture. · Finalize the Brand Vault data model, workflow design, and 3-round roadmap.	Complete architectural blueprint and clear product specification ready for build execution.
Round 2: The Build	· Build the core working MVP: Brand Vault Manager, **Shield Your Loom → LoomShield** Multi-Format Reviewer (text, `.html/.docx/.md`, visuals), 6-Pillar Dual Local Engine, Explainable Citation Cards, Surgical Diff Rewriter, and Role-Based Review Workflow. · Name the **three substantial improvements** to be delivered in Round 3.	End-to-end functional product running 100% locally with zero API keys required, plus clearly defined Round 3 targets.
Round 3: The Finish	Deliver the three major improvements named in Round 2: · **Improvement 1 — Two-Tier Hybrid Engine Pushed into Guideline Ambiguity Radar & Automated Clarity Rewriter:** Detect vague qualifiers and high reviewer override rates in brand guidelines, and automatically generate clearer, quantified rules with Do/Don't examples. · **Improvement 2 — Interactive Visual Clear-Space & WCAG Contrast Heatmap Inspector:** Add a 4×4 spatial image overlay showing region-by-region CIE ΔE color deviation, lockup clear-space margins, and text contrast ratios. · **Improvement 3 — Cryptographically Signed Pre-Publication Compliance Certificate:** Generate a SHA-256 governance sign-off bundle linking the approved content hash to the Brand Vault version.	Complete, stress-tested platform delivering all three pledged improvements, full documentation, and live demonstration.

9. The Expected Impact of My Solution

09 / IMPACT

- **Zero API Spend & Full Data Privacy:** Eliminates recurring AI token fees while keeping confidential pre-release campaigns 100% on-device.
- **Faster Review Turnaround:** Catches terminology, claim, color palette, clear-space, and WCAG contrast errors before human reviewers inspect the draft.
- **Verifiable Factual Accuracy:** Grounds promotional statistics against the Verified Fact Sheet with clear source citations.
- **Self-Improving Guidelines:** Helps brand leads spot and rewrite confusing guidelines that cause repeated mistakes.

10. Future Scope After the Event

10 / FUTURE

- **Vector & Design Tool Plugins:** Direct plugins for Figma, Illustrator, Google Docs, Notion, and WordPress checking SVG lockup geometry and copy live.
- **CI/CD Marketing Gates:** Automated Git pull-request checks that audit landing page HTML, markdown docs, and SVG marks before deployment.
- **Multi-Lingual Brand Governance:** Local cross-lingual embeddings to verify terminology and claims across regional markets.
- **Video & Motion Storyboard Auditing:** Local transcript and frame inspection for social video campaigns.

11. Complete Team and Participant Details

11 / PARTICIPANT

Field	Participant Details	Responsible Design & Tool Disclosures
Project Name	**LoomShield**	· **Privacy & Confidentiality:** Local-first processing ensures confidential brand assets remain on-device. · **Human Oversight:** Suggestions never auto-overwrite creator work; every fix requires explicit acceptance, modification, or documented override. · **Planned Open-Source Stack:** Python (FastAPI, Pydantic, Scikit-Learn, NumPy, Pillow, SQLite) and TypeScript/React/HTML/CSS.
Team Name	`isaac.incognito007`	
Participant Name	**Isaac Shiloah Herbert V** (Solo Team — 1 Member)	
Email ID	`isaac.incognito007@gmail.com`	
Role	Solo Architect, AI/NLP Engineer, Full-Stack Developer & Designer	